

Curriculum



Fernando Lucatelli Nunes

Born: January 31, 1990 (São Paulo – Brazil)

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- Earned Ph.D. in Mathematics with *summa cum laude* from University of Coimbra (UC|UP joint PhD Program in Mathematics), in 2018.
 - ❖ Thesis: Pseudomonads and Descent
 - ❖ Keywords: Pseudomonads; Adjoint Triangles; Bicategories; Categorical Structures; Coherence.
 - ❖ TID: 101570422.
 - ❖ Supervisor: Professor Dr. Maria Manuel Clementino.

Current Academic Position:

- Since 01/11/2020: PhD candidate in Computer Sciences. Utrecht University. Utrecht, Netherlands.
 - ❖ PhD candidate in the *Intelligent Software Systems Division* at Utrecht University.
 - ❖ Research group: *Software Technology*.
 - ❖ Keywords: Programming languages; Computational effects.
 - ❖ Supervisor: Dr. Matthijs Vákár

Academic Positions:

- 12/2018-12/2020: Research Fellow, Junior Researcher. University of Coimbra. Coimbra, Portugal.
 - ❖ Member of the *Centre for Mathematics, University of Coimbra*.
 - ❖ Research Group: *Algebra, Logic and Topology*.
 - ❖ 03/2020-12/2020: Faculty Member of the University of Coimbra | University of Porto Joint PhD Program in Mathematics.
- 01/09/2018-03/02/2019: Invited Assistant Professor. University of Coimbra. Portugal.
 - ❖ Undergraduate Program in Mathematics.
 - ❖ Topology and Linear Analysis.
- 04/04/2018-24/12/2018: Postdoctoral Fellow. Centre for Mathematics, University of Coimbra. Portugal.
 - ❖ Research Group: *Algebra, Logic and Topology*.
- 11/10/2012-24/01/2018: PhD candidate in Mathematics. UC|UP joint PhD Program in Mathematics. University of Coimbra. Coimbra, Portugal.
 - ❖ Thesis: Pseudomonads and Descent.
 - ❖ Keywords: Pseudomonads; Adjoint Triangles; Bicategories; Categorical Structures; Coherence.
 - ❖ Supervisor: Professor Dr. Maria Manuel Clementino.
 - ❖ 18/11/2013-24/01/2018: PhD candidate at the *Centre for Mathematics, University of Coimbra*.
 - ❖ Research Group: *Algebra, Logic and Topology*.

Other Academic Positions:

- 04/11/2019-18/11/2019: Visiting Researcher. University of Cape Town. Cape Town, South Africa.
 - ❖ Keywords: Grothendieck descent theory; Janelidze-Galois theory.
- 07/05/2018-22/05/2018: Visiting Researcher. Institut de Recherche en Mathématique et Physique, Université catholique de Louvain. Louvain-la-Neuve, Belgium.
 - ❖ Keywords: Grothendieck descent theory; Janelidze-Galois theory; Lax idempotent pseudomonads.
 - ❖ Supported by the *Institut de Recherche en Mathématique et Physique* and by the *Center for Mathematics, University of Coimbra*.

Awards/Distinctions:

- 2021: Oberwolfach Leibniz Fellowship. Mathematisches Forschungsinstitut Oberwolfach.

Research Units:

- Software Technology Research Group, Intelligent Software Systems Division. Utrecht University. Utrecht, Netherlands.
 - Since 11/2020: PhD candidate at the Software Technology Group.
- Centre for Mathematics, University of Coimbra. Coimbra, Portugal.
 - Since 24/12/2018: Researcher Member of the CMUC.
 - 08/04/2018-24/12/2018: Postdoctoral Fellow at the CMUC.
 - 18/11/2013-24/01/2018: PhD candidate at the CMUC.
- University of Coimbra-University of Porto Joint PhD Program in Mathematics.
 - 03/2020-12/2021: Faculty Member.

Publications:

1. **PhD thesis:** *Pseudomonads and Descent*. University of Coimbra. 2018.

2. Articles:

Published articles:

- Lucatelli Nunes, Fernando. Pseudoalgebras and non-canonical isomorphisms. *Applied Categorical Structures*, v. 27, n 1, pp 55-63. 2019.
- Lucatelli Nunes, Fernando. On lifting of biadjoints and lax algebras. *Categories and General Algebraic Structures with Applications*, v. 9, n. 1, pp 29-58. 2018.
- Lucatelli Nunes, Fernando. Pseudo-Kan Extensions and Descent Theory. *Theory and Applications of Categories*, v. 33, no. 15, pp 390-444. 2018.
- Lucatelli Nunes, Fernando. On biadjoint triangles. *Theory and Applications of Categories*, v. 31, n. 9, p. 217-256. 2016.
- Lucatelli Nunes, Fernando. Espaços não reversíveis. *Revista Matemática Universitária*, v. 48, n. 49, p. 22-26. 2012.

Submitted/under revision:

- Lucatelli Nunes, Fernando. Freely generated n-categories, coinserter and presentations of low dimensional categories. arXiv:1704.04474. 2017.
- Lucatelli Nunes, Fernando. Semantic Factorization and Descent. arXiv:1902.01225. 2019.
- Lucatelli Nunes, Fernando. Descent Data and Absolute Kan Extensions. arXiv:1906.00517. 2019.
- Clementino, Maria Manuel and Lucatelli Nunes, Fernando. Lax comma 2-categories and admissible 2-functors. arXiv:2002.03132. 2020.

3. Other Publications:

Annals/Proceedings of Conferences:

- Lucatelli Nunes, Fernando; Diniz, P.C.; MENDES, V. M.; MIRANDA, W. D. S. A. *Teoria dos Jogos: Equilíbrio de Nash e Jogos Evolucionários*. Annals of 64a Reunião Anual da SBPC. Brazilian Society for Scientific Progress. São Luís, MA, Brazil. 2012.
- Lucatelli Nunes, Fernando. *Teoria de Homotopia Moderna: CATEGORIAS MODELO*, Annals of 64a Reunião Anual da SBPC. Brazilian Society for Scientific Progress. São Luís, MA, Brazil. 2012.
- Lucatelli Nunes, Fernando. *Teoria das Categorias: Adjunções, Colimites e Limites*. Annals of 64a Reunião Anual da SBPC. Brazilian Society for Scientific Progress, São Luís, MA, Brazil. 2012.
- Lucatelli Nunes, Fernando. *CATEGORIAS MODELO: COLIMITES DE HOMOTOPIA*. Annals of 64a Reunião Anual da SBPC. Brazilian Society for Scientific Progress, São Luís, MA, Brazil. 2012.
- Lucatelli Nunes, Fernando; Diniz, P.C.; MIRANDA, W. D. S. A.; Neto, L.M.A. *Um convite à topologia: demonstrações topológicas de resultados clássicos*. Annals of 64a Reunião Anual da SBPC. Brazilian Society for Scientific Progress. São Luís, MA, Brazil. 2012.
- Lucatelli Nunes, Fernando. *Grupóides na Topologia*. Annals of 63a Reunião Anual da SBPC. Brazilian Society for Scientific Progress. Goiânia, GO, Brazil. 2011.
- Lucatelli Nunes, Fernando. *Dinâmica Topológica e Aplicações à Teoria dos Números*. Annals of 63a Reunião Anual da SBPC. Brazilian Society for Scientific Progress. Goiânia, GO, Brazil. 2011.

Monographs:

- Lucatelli Nunes, Fernando. *Elementos de Topologia para Sistemas Dinâmicos* ("Basic concepts of topology for dynamical systems"). UnB: (Monograph in portuguese), 117 pages. 2010.
<https://liedinamica.files.wordpress.com/2019/07/topologia2010.pdf>
- Lucatelli Nunes, Fernando. *Dinâmica Topológica e Aplicações à Teoria dos Números* ("Topological Dynamics and Applications to number theory"). UnB: (Monograph in portuguese), 54 pages. 2010.
<https://teoriadelie.files.wordpress.com/2010/11/dinc3a2mica-topolc3b3gica-e-aplicac3a7c3b5es-c3a0-teoria-dos-nc3bameros1.pdf>

Talks:

Conferences:

- Eilenberg-Moore, Kleisli and descent factorization. *One-day conference on Categorical Structures in Algebra, Topology and Logic*. Centre for Mathematical Structures, Stellenbosch University. South Africa. 2019.
- Lax comma 2-categories. XII Portuguese Category Seminar. University of Coimbra, Portugal. 2019.
<http://mat.uc.pt/~categ/events/september2019.html>
- Descent and Monadicity. CT 2019, *International Conference on Category Theory 2019*. University of Edinburgh. Scotland. 2019.
<http://conferences.inf.ed.ac.uk/ct2019/>
- Aspects of Descent via Bilimits. *Plenary talk*. CT 2018, *International Conference on Category Theory 2018*. University of Azores. Portugal. 2018.
<http://www.mat.uc.pt/~ct2018/>
- Biadjoint Triangles and Applications. *103rd Peripatetic Seminar on Sheaves and Logic (PSSL)*. Masaryk University. Brno. Czech Republic. 2018.
<https://www.math.muni.cz/~loregianf/PSSL103/schedule.php>
- Non-canonical isomorphisms. *102nd Peripatetic Seminar on Sheaves and Logic (PSSL)*. Universidade de Santiago de Compostela. Spain. 2018.
<https://www.usc.es/regaca/pssl102/>
- Biadjoint Triangles. *5th Workshop on Categorical Methods in Non-Abelian Algebra*. Université catholique de Louvain. Louvain-la-Neuve, Belgium. 2017.
<http://www.mat.uc.pt/~mmc/mcana/MCANAfifthworkshop.html>
- Freely generated n-categories and coinserters. XI Portuguese Category Seminar, University of Coimbra. Portugal. 2017.
<http://www.mat.uc.pt/~categ/events/programa2017.html>
- Bilimits Commutativity and Descent. *4th Workshop on Categorical Methods in Non-Abelian Algebra*. University of Coimbra. Portugal. 2016.
http://www.mat.uc.pt/~mmc/mcana/MCANAfourthworkshop_programme.html
- Kan extensions and descent theory. CT 2015, *International Conference on Category Theory 2015*. University of Aveiro. Portugal. 2015.
<http://sweet.ua.pt/dirk/ct2015/>

Seminars:

- Functorial Semantics and Descent II. *Algebra, Logic and Topology Seminar*. Centre for Mathematics, University of Coimbra. Portugal. 2019.
<http://www.mat.uc.pt/~categ/seminars/>
- On monadicity and descent. *(Informal) Seminar of the Category Theory Group at the University of Coimbra*. University of Coimbra. Portugal. 2019.
<http://www.mat.uc.pt/~categ/seminars/>
- Functorial Semantics and Descent. *Algebra, Logic and Topology Seminar*. Centre for Mathematics, University of Coimbra. Portugal. 2018.
<http://www.mat.uc.pt/~categ/seminars/>
- Towards semi left exactness and Janelidze-Galois within the lax idempotent context. *Algebra, Logic and Topology Seminar*. Centre for Mathematics, University of Coimbra.

Portugal. 2018.

<http://www.mat.uc.pt/~categ/seminars/>

- The deficiency of categories. *Algebra, Logic and Topology Seminar*. Centre for Mathematics, University of Coimbra. Portugal. 2016.
<http://www.mat.uc.pt/~categ/seminars/>
- Biadjoint triangles, descent and coherence. *Algebra, Logic and Topology Seminar*, CMUC. University of Coimbra. Portugal. 2015.
<http://www.mat.uc.pt/~categ/seminars/>
- Dinâmica Topológica e Aplicações à Teoria dos Números. *Math Seminar*. University of Brasília. Brazil. 2010.
- Demonstração Dinâmica do Teorema de Van der Waerden. *Math Seminar*. University of Brasília. Brazil. 2010.

PhD seminars (for PhD students):

- Kan construction of adjunctions. *Research Seminar Program*. University of Porto. Porto, Portugal. 2016.
<https://sites.google.com/view/rspucupphdmath>
- Homotopy Excision. *Research Seminar Program*. University of Coimbra. Portugal. 2016.
<https://sites.google.com/view/rspucupphdmath>
- Every prespectrum represents a homology theory. *Joint PhD Program UC-UP Seminar*. Universidade de Coimbra. Portugal. 2013.
<http://www.mat.uc.pt/~categ/seminars/>

Posters:

- Categorias Modelo: COLIMITES DE HOMOTOPIA. *64ª Reunião Anual da SBPC*. Sociedade Brasileira para o Progresso da Ciência. Universidade Federal do Maranhão. Brazil. 2012.
- Teoria de Homotopia Moderna: CATEGORIAS MODELO. *64ª Reunião Anual da SBPC*. Sociedade Brasileira para o Progresso da Ciência. Universidade Federal do Maranhão. Brazil. 2012.
- Grupóides na Topologia. *63ª Reunião Anual da SBPC*. Sociedade Brasileira para o Progresso da Ciência. Universidade Federal de Goiás. Brazil. 2011.
- Dinâmica Topológica e Aplicações à Teoria dos Números. *63ª Reunião Anual da SBPC*. Sociedade Brasileira para o Progresso da Ciência. Universidade Federal de Goiás. Brazil. 2011.
- Dinâmica Topológica e Aplicações à Teoria dos Números. *Escola e Workshop de Teoria de Lie*. UNICAMP. Campinas, Brazil. 2010.

Teaching:

PhD level:

- 01/2020-07/2020: University of Coimbra *and* University of Porto. Portugal.
 - ❖ Joint PhD Program in Mathematics University of Coimbra-University of Porto.
 - ❖ Categories in Algebra and Topology (with Professor Dr. Maria Manuel Clementino).

Undergraduate level:

- 09/2018-02/2019: University of Coimbra. Portugal.
 - ❖ Undergraduate Program in Mathematics (UC).
 - ❖ Invited Assistant Professor.
 - ❖ Topology and Linear Analysis (with Professor Dr. Maria Manuel Clementino).

Teaching assistant/tutor:

- 01/2012-06/2012: Universidade de Brasília. Brazil.
 - ❖ Undergraduate teaching assistant
 - ❖ Linear Algebra.
- 09/2010-04/2011: Universidade de Brasília. Brazil.
 - ❖ Undergraduate teaching assistant
 - ❖ Analysis of $\mathbb{R}^n$ (Professor Dr. Elvies A. de Barros e Silva).
- 02/2010-09/2010: Universidade de Brasília. Brazil.
 - ❖ Undergraduate teaching assistant
 - ❖ Algebra (Professor Lineu Neto).
- 06/2009-12/2009: Universidade de Brasília. Brazil.
 - ❖ Undergraduate teaching assistant
 - ❖ Real Analysis (Professor Dr. José Alfredo Silva).
- 06/2008-12/2008: Universidade de Brasília. Brazil.
 - ❖ Undergraduate teaching assistant
 - ❖ Differential and Integral Calculus.

Supervision:

PhD student:

- Since 01/2020: Joint UC-UP PhD program in Mathematics. PhD candidate: Rui *Rodrigues de Abreu Fernandes Prezado*, ongoing, supervision with Prof. Dr. Maria Manuel Clementino. Topic: Some Aspects of Descent Theory and Applications

Reviewer and Referee Experience:

- Since 2016: Reviewer of *zbMATH* (lucatelli-nunes.fernando).
- Since 2016: Reviewer of *Mathematical Reviews* (1158728).
- Since 2018: Referee of the journal *Applied Categorical Structures*.
- Since 2018: Referee of the journal *Topology and its Applications*.

Organization of Seminars:

- Since 12/2019: Organizer of the *Algebra, Logic and Topology Seminar* of the Center for Mathematics, University of Coimbra. Portugal.
(<https://cmuc.mat.uc.pt/rdonweb/event/ppseminarhistory.do>)

(Co)organization of Events:

- *XII Portuguese Category Seminar*. University of Coimbra, Portugal. 2019.
- *Workshop on Algebra, Logic and Topology*. University of Coimbra, Portugal. 2018.
- *Workshop: Logic for Children*, UNILOG 2018. Vichy, France. 2018.

Collaborated with (outreach activities):

- International Mathematical Olympiad. (Portuguese) IMO commissioner. 2020.
- 09/2017-12/2020: Delfos (extension project: “Escola Delfos”) for students interested in Mathematics in the *Math Olympiads’ level* (<https://www.uc.pt/fctuc/dmat/delfos> and <http://www.uc.pt/fctuc/dmat/delfos/estagios>).
- 2017: *Mostra de Ciências* (Science communication, organized by Prof. Dr. Jorge Picado)
- <https://liedinamica.wordpress.com/>
- Grupos de Matrizes <https://liedinamica.wordpress.com/category/wikis/grupos-de-matrizes/>

Mathoverflow and Mathstackexchange:

- Mathoverflow: <https://mathoverflow.net/users/18017/fernando>
- Math.stackexchange: <https://math.stackexchange.com/users/71821/fernando>

Mathematical Genealogy:

- <https://genealogy.math.ndsu.nodak.edu/id.php?id=228539>

Additional Information at:

- Personal webpage: <http://www.mat.uc.pt/~fernandolucatelli/>
- arXiv: https://arxiv.org/a/lucatellinunes_f_1.html
- Google Scholar: https://scholar.google.fr/citations?user=QoW_v8QAAAAJ&hl=pt-BR
- Research Gate: https://www.researchgate.net/profile/Fernando_Lucatelli_Nunes
- MR author ID: 1158728
- ORCID: 0000-0002-1817-2797
- ResearcherID: A-6714-2017

Participation in events:

- One-day conference on Categorical Structures in Algebra, Topology and Logic. Stellenbosch University. South Africa. 2019.
- XII Portuguese Category Seminar. University of Coimbra. Portugal. 2019.
- Descent and chromatic homotopy theory. University of Strasbourg. France. 2019.
- CT 2019, International Conference on Category Theory 2019. University of Edinburgh. Scotland. 2019.
- ACT 2019, Applied Category Theory Conference 2019. University of Oxford. England. 2019.
- Portuguese Workshop on Semigroups. University of Coimbra. Portugal. 2019.
- Aspects of Higher Representation Theory, Mini-Course and Conference. Vrije Universiteit Brussel. Belgium. 2019.
- Topology Seminar. University of Aveiro. Portugal. 2018.
- Workshop on Algebra, Logic and Topology. University of Coimbra. Portugal. 2018.
- CT 2018, International Conference on Category Theory 2018. University of Azores. Portugal. 2018.
- 103rd Peripatetic Seminar on Sheaves and Logic (103rd PSSL). Masaryk University. Brno, Czech Republic. 2018.
- 102nd Peripatetic Seminar on Sheaves and Logic (102nd PSSL). Universidade de Santiago de Compostela. Spain. 2018.
- 3rd CMUC Workshop. University of Coimbra. Portugal. 2017.
- 5th Workshop on Categorical Methods in Non-Abelian Algebra. Université catholique de Louvain. Belgium. 2017.
- XI Portuguese Category Seminar. University of Coimbra. Portugal. 2017.
- 100th Peripatetic Seminar on Sheaves and Logic. University of Cambridge. England. 2016.
- 4th Workshop on Categorical Methods in Non-Abelian Algebra. University of Coimbra, Portugal. 2016.
- Workshop on Dualities. University of Coimbra, Portugal. 2016.
- 3rd Workshop on Categorical Methods in Non-Abelian Algebra. Université catholique de Louvain. Belgium. 2015.
- 97th Peripatetic Seminar on Sheaves and Logic (97th PSSL). Université catholique de Louvain. Louvain-la-Neuve, Belgium. 2015.
- AMS-EMS-SPM International Meeting. University of Porto, Portugal. 2015.
- CT 2015, International Conference on Category Theory 2015. University of Aveiro, Portugal. 2015.
- Mathematics, Culture and Creation. Fundação Francisco Manuel dos Santos. University of Coimbra. 2015.
- Workshop in Category Theory and Algebraic Topology. UCL, Louvain-la-Neuve, Belgium. 2015.
- CT 2014, International Conference on Category Theory 2014. University of Cambridge. England. 2014.
- Workshop on Categorical Methods in Algebra and Topology, in honor of Manuela Sobral. University of Coimbra. Portugal. 2014.
- X Portuguese Category Seminar. University of Coimbra. Portugal. 2014.

- 1st Workshop on Categorical Methods in Non-Abelian Algebra. University of Coimbra. Portugal. 2013.
- IX Portuguese Category Seminar. University of Coimbra. Portugal. 2013.
- 64ª Reunião Anual da SBPC – 64th Annual Meeting of the Brazilian Society for Scientific Progress. Universidade Federal do Maranhão. São Luís, Maranhão, Brazil. 2012.
- 63ª Reunião Anual da SBPC – 63th Annual Meeting of the Brazilian Society for Scientific Progress. Universidade Federal de Goiás. Goiânia, Brazil. 2011.
- Escola e Workshop de Teoria de Lie – School and Workshop on Lie Theory. UNICAMP – Universidade Estadual de Campinas. São-Paulo, Brazil. 2010.
- XVII Encontro Brasileiro de Topologia – XVII Brazilian Topology Meeting. Pontifícia Universidade Católica do Rio de Janeiro. Brazil. 2010.
- XXI Escola de Álgebra – XXI Brazilian Algebra Meeting. University of Brasília. Brazil. 2010.